



QUESTIONS

10 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate $K (0 \leq K < N)$, show

original → rotated → reversal steps visually, compute the sum difference, and support multiple rotations.

Sample Input/Output:

Enter 10 numbers: 1 2 3 4 5 6 7 8 9 10

Rotate Left (L)/Right (R): L

By positions: 3

Original: [1 2 3 4 5 6 7 8 9 10]

Step1: Reverse 0-2 → [3 2 1 4 5 6 7 8 9 10]

Step2: Reverse 3-9 → [3 2 1 10 9 8 7 6 5 4]

Step3: Reverse all → [4 5 6 7 8 9 10 1 2 3]

Sum difference: 0"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 #define N 10
4
5 void reverse(int arr[], int start, int end)
6 {
7     while (start < end)
8     {
9         int temp = arr[start];
10        arr[start] = arr[end];
11        arr[end] = temp;
12        start++;
13        end--;
14    }
15 }
16
17 void printArray(int arr[])
18 {
19     int i;
20     printf("[");
21     for (i = 0; i < N; i++)
```

OUTPUT

```
Enter 10 numbers: Rotate Left (L)/Right (R): By positions:
Original: [0 0 0 0 -1854990352 455 62133513 32758 1 0]
Invalid choice
```

INPUT

Enter 10 numbers: